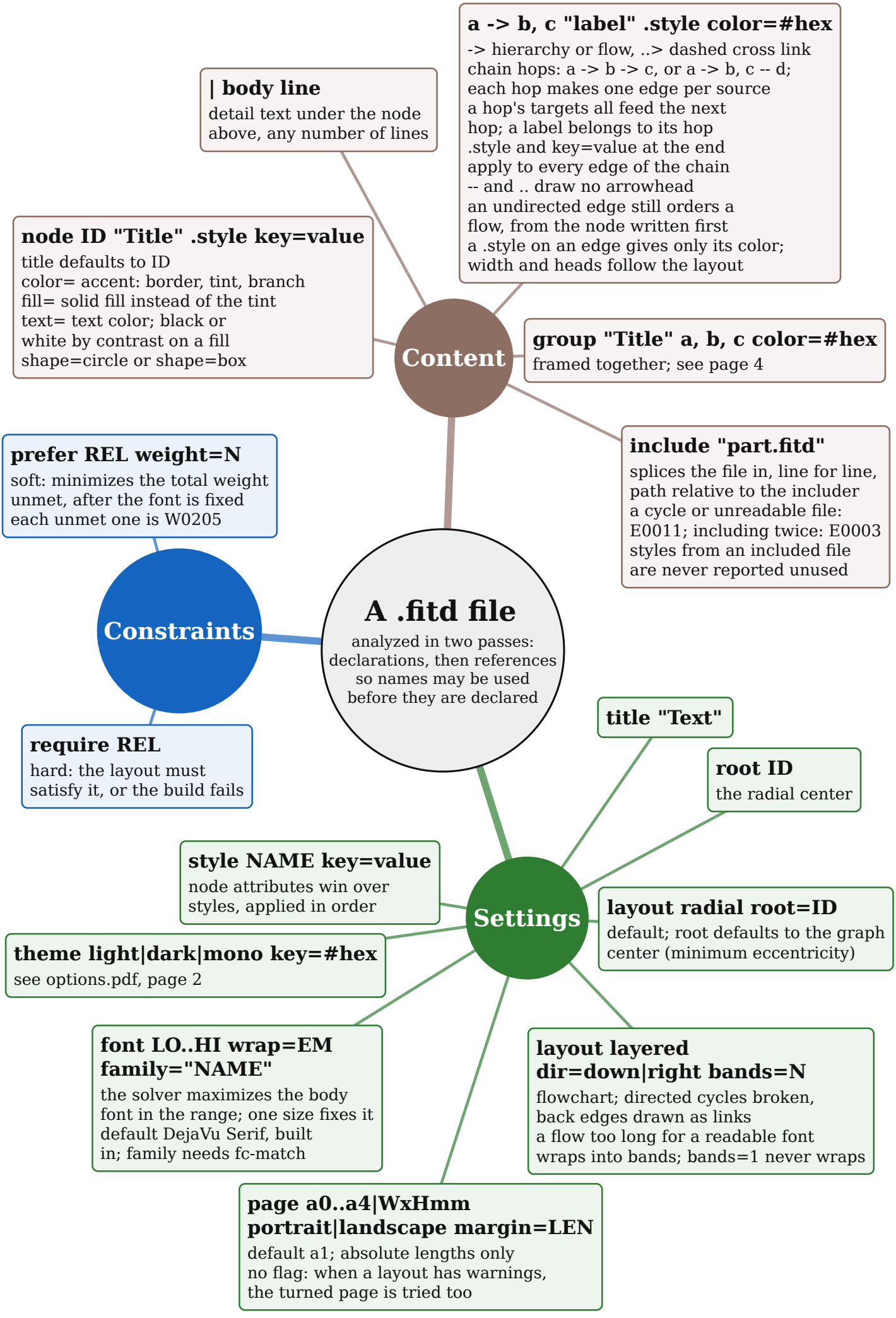
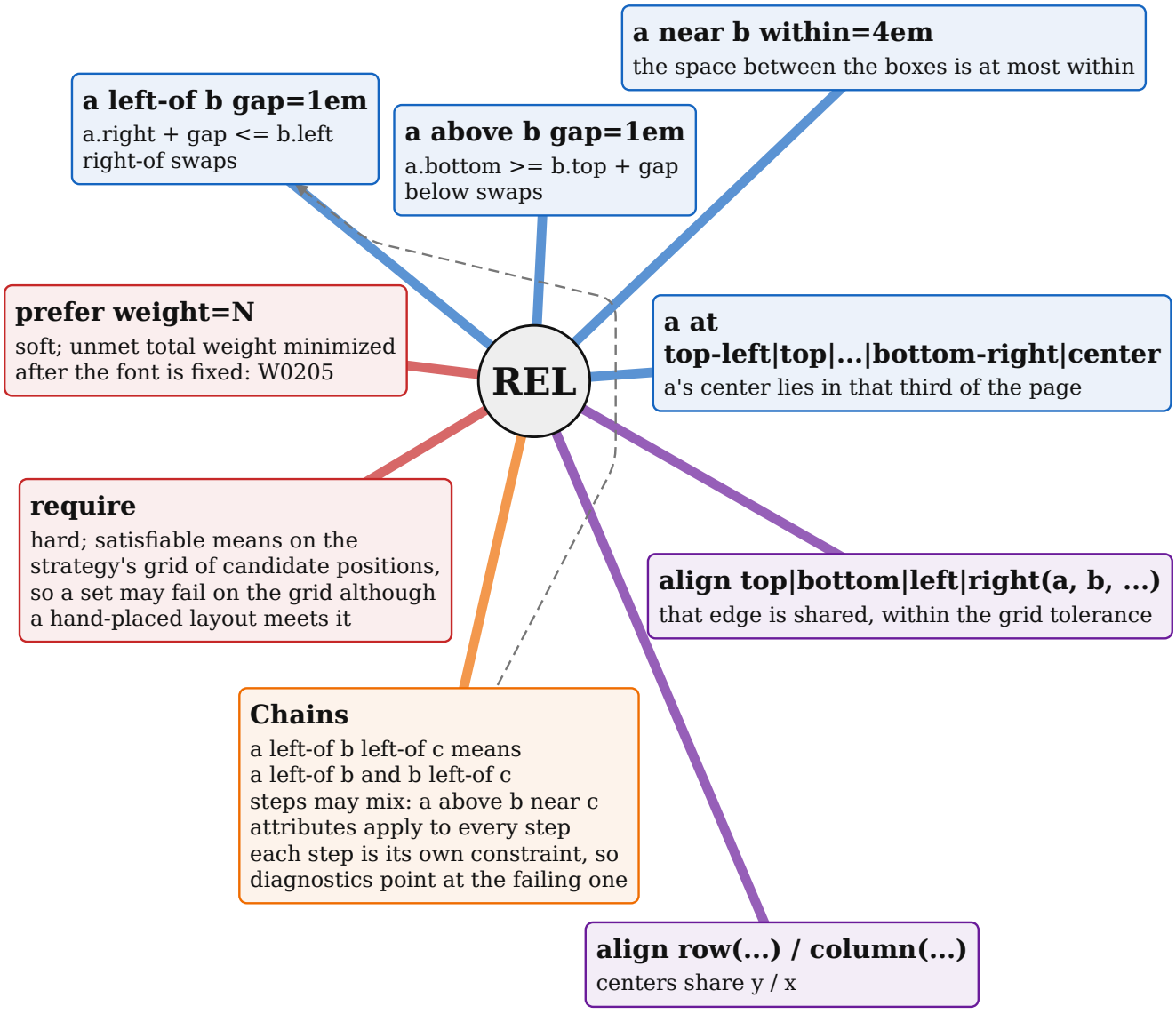
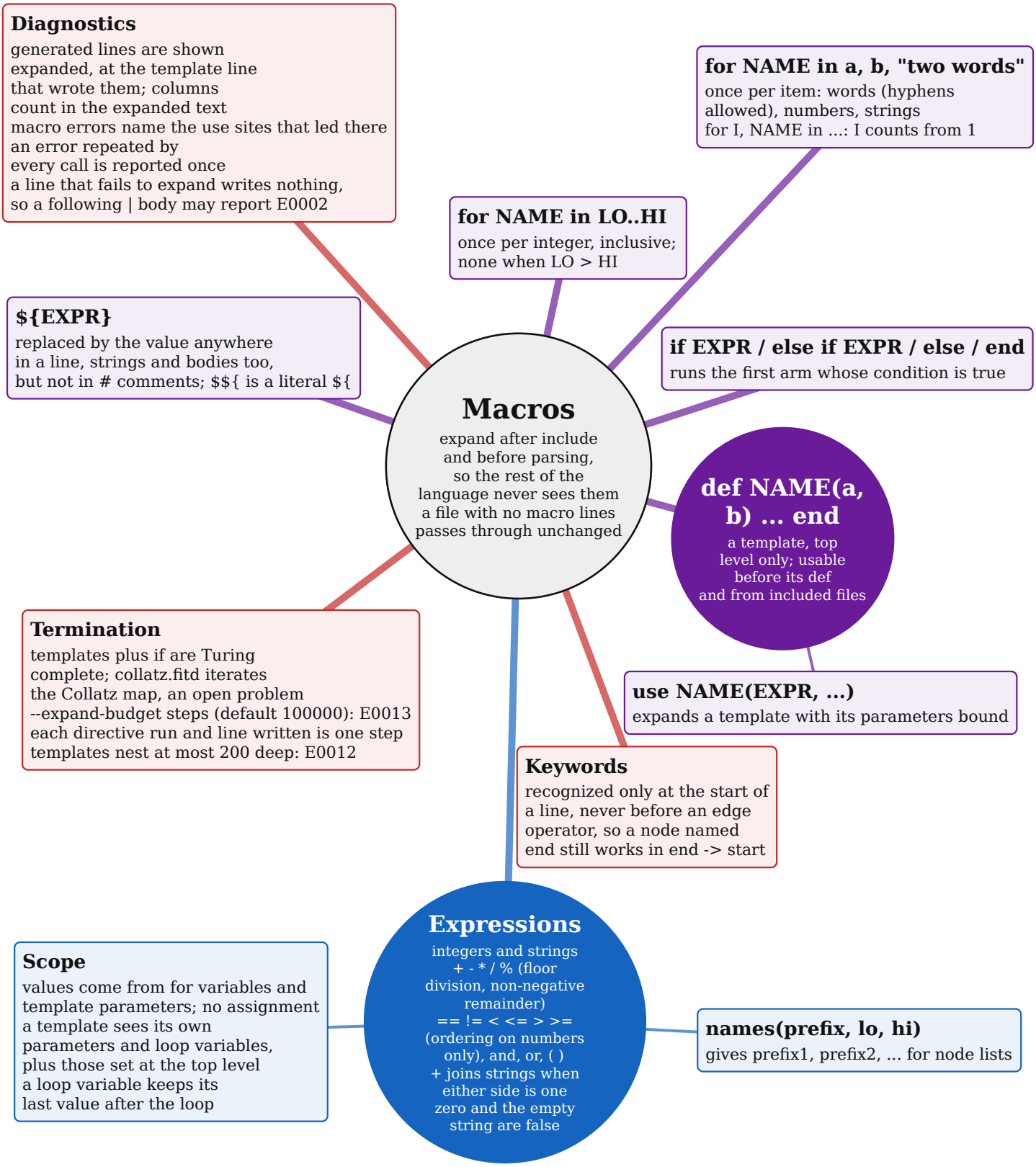
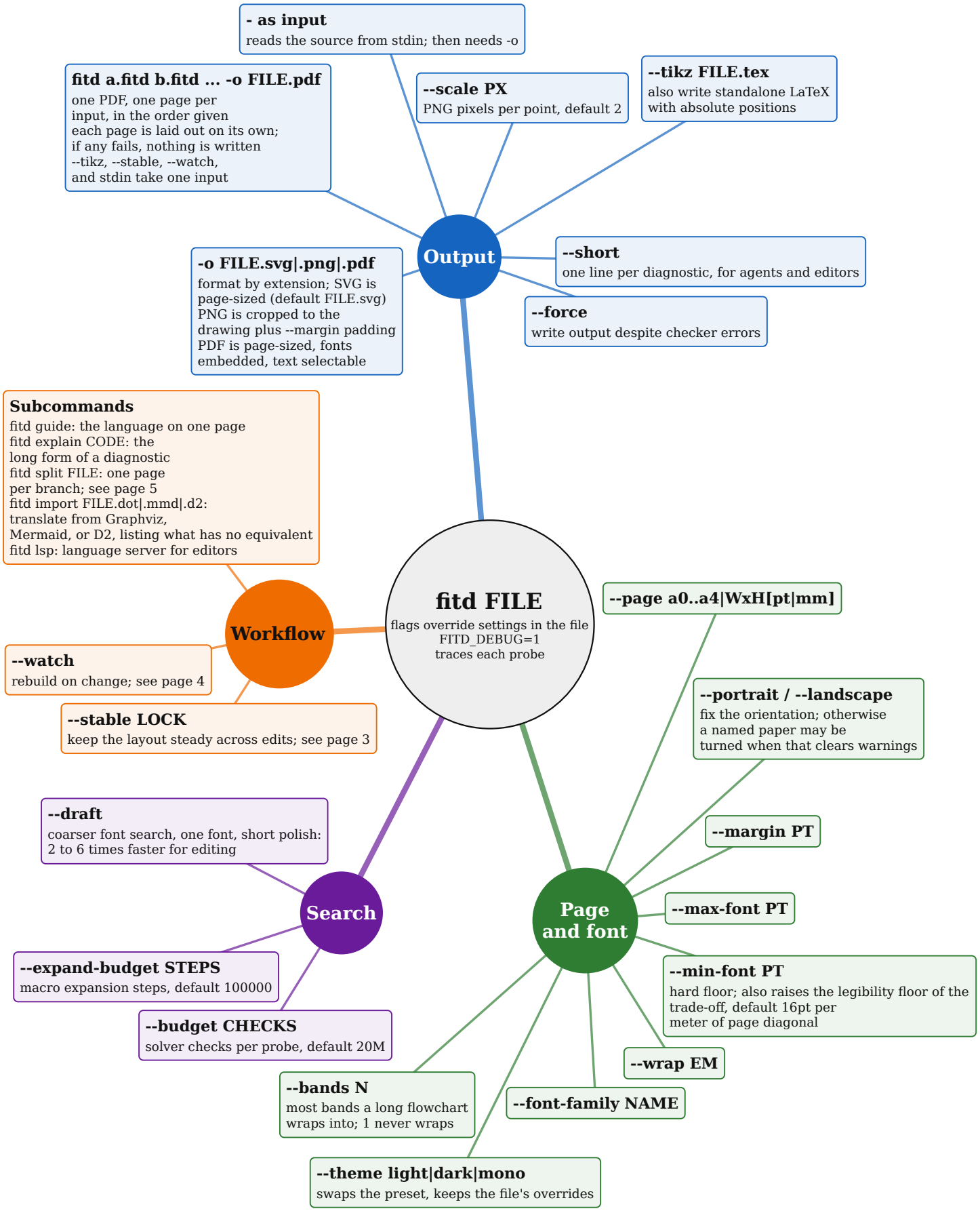

A .fitd file
analyzed in two passes: declarations, then references so names may be used before they are declared
Content
a -> b, c "label" .style color=#hex
-> hierarchy or flow, ..> dashed cross link
chain hops: a -> b -> c, or a -> b, c -- d;
each hop makes one edge per source
a hop's targets all feed the next hop; a label belongs to its hop
.style and key=value at the end apply to every edge of the chain
-- and .. draw no arrowhead
an undirected edge still orders a flow, from the node written first
a .style on an edge gives only its color; width and heads follow the layout
| body line
detail text under the node above, any number of lines
node ID "Title" .style key=value
title defaults to ID
color= accent: border, tint, branch
fill= solid fill instead of the tint
text= text color; black or white by contrast on a fill
shape=circle or shape=box
group "Title" a, b, c color=#hex
framed together; see page 4
include "part.fitd"
splices the file in, line for line, path relative to the includer
a cycle or unreadable file: E0011; including twice: E0003
styles from an included file are never reported unused
Constraints
prefer REL weight=N
soft: minimizes the total weight unmet, after the font is fixed
each unmet one is W0205
require REL
hard: the layout must satisfy it, or the build fails
Settings
title "Text"
root ID
the radial center
style NAME key=value
node attributes win over styles, applied in order
theme light|dark|mono key=#hex
see options.pdf, page 2
layout radial root=ID
default; root defaults to the graph center (minimum eccentricity)
font LO..HI wrap=EM family="NAME"
the solver maximizes the body font in the range; one size fixes it
default DejaVu Serif, built in; family needs fc-match
layout layered dir=down|right bands=N
flowchart; directed cycles broken, back edges drawn as links
a flow too long for a readable font wraps into bands; bands=1 never wraps
page a0..a4|WxHmm portrait|landscape margin=LEN
default a1; absolute lengths only
no flag: when a layout has warnings, the turned page is tried too


a near b within=4em
the space between the boxes is at most within
a left-of b gap=1em
a.right + gap <= b.left
right-of swaps
a above b gap=1em
a.bottom >= b.top + gap
below swaps
prefer weight=N
soft; unmet total weight minimized
after the font is fixed: W0205
REL
a at
top-left|top|...|bottom-right|center
a's center lies in that third of the page
require
hard; satisfiable means on the
strategy's grid of candidate positions,
so a set may fail on the grid although
a hand-placed layout meets it
align top|bottom|left|right(a, b, ...)
that edge is shared, within the grid tolerance
Chains
a left-of b left-of c means
a left-of b and b left-of c
steps may mix: a above b near c
attributes apply to every step
each step is its own constraint, so
diagnostics point at the failing one
align row(...) / column(...)
centers share y / x


Diagnostics
generated lines are shown
expanded, at the template line
that wrote them; columns
count in the expanded text
macro errors name the use sites that led there
an error repeated by
every call is reported once
a line that fails to expand writes nothing,
so a following | body may report E0002
for NAME in a, b, "two words"
once per item: words (hyphens
allowed), numbers, strings
for I, NAME in ...: I counts from 1
for NAME in LO..HI
once per integer, inclusive;
none when LO > HI
${EXPR}
replaced by the value anywhere
in a line, strings and bodies too,
but not in # comments; $${ is a literal ${
if EXPR / else if EXPR / else / end
runs the first arm whose condition is true
Macros
expand after include
and before parsing,
so the rest of the
language never sees them
a file with no macro lines
passes through unchanged
def NAME(a, b) ... end
a template, top
level only; usable
before its def
and from included files
Termination
templates plus if are Turing
complete; collatz.fitd iterates
the Collatz map, an open problem
--expand-budget steps (default 100000): E0013
each directive run and line written is one step
templates nest at most 200 deep: E0012
use NAME(EXPR, ...)
expands a template with its parameters bound
Keywords
recognized only at the start of
a line, never before an edge
operator, so a node named
end still works in end -> start
Expressions
integers and strings
+ - * / % (floor
division, non-negative
remainder)
== != < <= > >=
(ordering on numbers
only), and, or, ( )
+ joins strings when
either side is one
zero and the empty
string are false
Scope
values come from for variables and
template parameters; no assignment
a template sees its own
parameters and loop variables,
plus those set at the top level
a loop variable keeps its
last value after the loop
names(prefix, lo, hi)
gives prefix1, prefix2, ... for node lists


fitd FILE
flags override settings in the file
FITD_DEBUG=1
traces each probe
Output
- as input
reads the source from stdin; then needs -o
fitd a.fitd b.fitd ... -o FILE.pdf
one PDF, one page per
input, in the order given
each page is laid out on its own;
if any fails, nothing is written
--tikz, --stable, --watch,
and stdin take one input
--scale PX
PNG pixels per point, default 2
--tikz FILE.tex
also write standalone LaTeX
with absolute positions
-o FILE.svg|.png|.pdf
format by extension; SVG is
page-sized (default FILE.svg)
PNG is cropped to the
drawing plus --margin padding
PDF is page-sized, fonts
embedded, text selectable
--short
one line per diagnostic, for agents and editors
--force
write output despite checker errors
Workflow
Subcommands
fitd guide: the language on one page
fitd explain CODE: the
long form of a diagnostic
fitd split FILE: one page
per branch; see page 5
fitd import FILE.dot|.mmd|.d2:
translate from Graphviz,
Mermaid, or D2, listing what has no equivalent
fitd lsp: language server for editors
--watch
rebuild on change; see page 4
--stable LOCK
keep the layout steady across edits; see page 3
Search
--draft
coarser font search, one font, short polish:
2 to 6 times faster for editing
--expand-budget STEPS
macro expansion steps, default 100000
--budget CHECKS
solver checks per probe, default 20M
Page
and font
--page a0..a4|WxH[pt|mm]
--portrait / --landscape
fix the orientation; otherwise
a named paper may be
turned when that clears warnings
--margin PT
--max-font PT
--min-font PT
hard floor; also raises the legibility floor of the
trade-off, default 16pt per
meter of page diagonal
--wrap EM
--font-family NAME
--bands N
most bands a long flowchart
wraps into; 1 never wraps
--theme light|dark|mono
swaps the preset, keeps the file's overrides